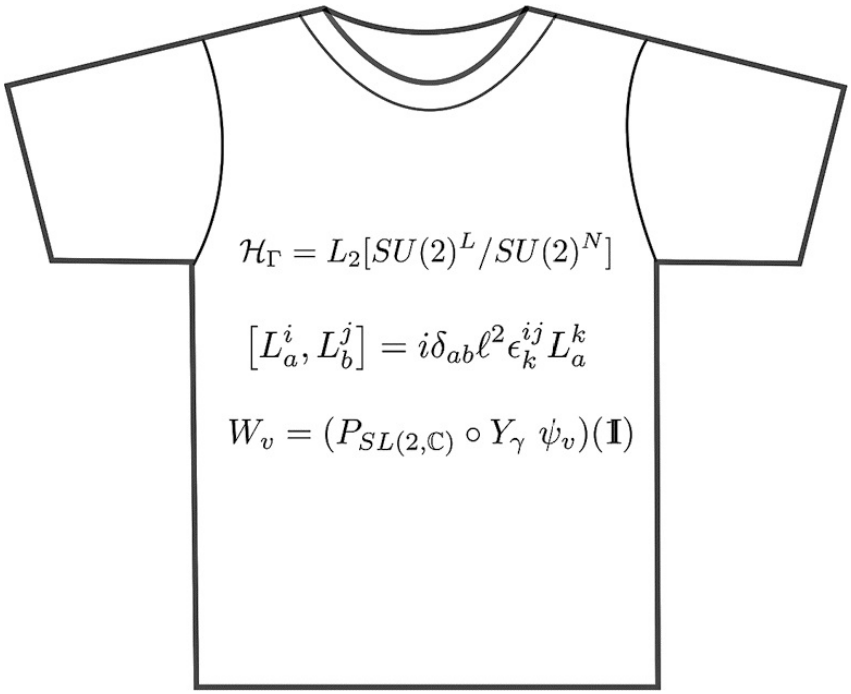


Figure 7.7 The equations of loop quantum gravity, summarized on a T-shirt.

These equations[fn44] are the mathematical version of the picture of the world I have given in the last two chapters. We are not at all sure if they are the correct equations – but, in my opinion, they are the best account of quantum gravity we have at present.

Space is a spin network whose nodes represent its elementary grains, and whose links describe their proximity relations. Spacetime is generated by processes in which these spin networks transform into one another, and these processes are described by sums over spinfoams. A spinfoam represents a history of a spin network, hence a granular spacetime where the nodes of the graph combine and separate.

This microscopic swarming of quanta, which generates space and time, underlies the calm appearance of the macroscopic reality surrounding us. Every cubic centimetre of space, and every second that passes, is the result of this dancing foam of extremely small quanta.

What is the world made of?

The backdrop of space has disappeared, time has disappeared, classic particles have disappeared, along with the classic fields. So what is the world made of?

The answer now is simple: the particles are quanta of quantum fields; light is formed by quanta of a field; space is nothing more than a field, which is also made of quanta; and time emerges from the processes of this same field. In other words, the world is made entirely from quantum fields (figure 7.8).